

EXPERIMENT-20 Planets Database

Aim

To create a **Prolog database of planets** and retrieve information about planets.

Objectives

- To store planetary information as Prolog facts.
- To query the database.
- To understand facts and predicates.

Algorithm

1. Define planet facts.
2. Store planet name, order and type.
3. Query the database.
4. Retrieve matching information.
5. Display the result.

Flowchart

START

|

v

Create planet

facts

|

v

Enter query

|

v

Search database

|

v

Match found?

/ \

Yes No

|

|

v

v

Display Fail

result

|

v

STOP

Prolog Program

% Planets Database

planet(mercury, 1, terrestrial).

planet(venus, 2, terrestrial).

planet(earth, 3, terrestrial).

planet(mars, 4, terrestrial).

planet(jupiter, 5, gas_giant).

planet(saturn, 6, gas_giant).

planet(uranus, 7, ice_giant).

planet(neptune, 8, ice_giant).

Query

?- planet(earth, Order, Type).

Output

```
:- initialization(main).
% Planets Database

planet(mercury, 1, terrestrial).
planet(venus, 2, terrestrial).
planet(earth, 3, terrestrial).
planet(mars, 4, terrestrial).
planet(jupiter, 5, gas_giant).
planet(saturn, 6, gas_giant).
planet(uranus, 7, ice_giant).
planet(neptune, 8, ice_giant).

% Display all planets
main :-
    planet(Name, Number, Type),
    write(Name),
    write(' - Position: '),
    write(Number),
    write(' - Type: '),
    write(Type),
    nl,
    fail.

main.

:- initialization(main).
```

```
Input for the program ( Optional )

Output:
mercury - Position: 1 - Type: terrestrial
venus - Position: 2 - Type: terrestrial
earth - Position: 3 - Type: terrestrial
mars - Position: 4 - Type: terrestrial
jupiter - Position: 5 - Type: gas_giant
saturn - Position: 6 - Type: gas_giant
uranus - Position: 7 - Type: ice_giant
neptune - Position: 8 - Type: ice_giant
mercury - Position: 1 - Type: terrestrial
venus - Position: 2 - Type: terrestrial
earth - Position: 3 - Type: terrestrial
mars - Position: 4 - Type: terrestrial
jupiter - Position: 5 - Type: gas_giant
saturn - Position: 6 - Type: gas_giant
uranus - Position: 7 - Type: ice_giant
neptune - Position: 8 - Type: ice_giant
```

Result

The planets database was successfully created and queried using Prolog.

Conclusion

Thus, a Prolog program was successfully implemented to represent and retrieve planetary information.